

ControlPlane.ai

Finding it first — not finding out.

A real-time risk-checking layer that sits on top of any AI deployment and evaluates every response across three dimensions — **Performance** (is it grounded / confidently wrong?), **Cost** (is it burning compute or rework?), and **Responsibility** (is it biased, unsafe, or leaking data?) — before it reaches a user or downstream system.

This repository contains a working prototype of the core detection-and-decision mechanism, submitted for Round 2 of the Accenture Innovation Challenge 2026.

Live demo

Open `index.html` in any browser — no install, no API key, no backend required. Everything runs client-side so it's safe to demo offline.

1. **Run a live check** — paste an AI response (and optionally a source document) and watch the three detectors score it in real time.
2. **Run the batch demo** — replays 18 simulated interactions across all three use cases (customer chatbot, internal copilot, regulated decision-support), including PII leaks, hallucinated claims, cost outliers, borderline/ambiguous cases, and one bias case, to show system behavior at scale.
3. **Check the confidence indicator** — every check reports a confidence percentage alongside its risk score, separate from the risk level itself. Confidence drops when there's no source to verify against, when the score sits right on a decision-tier boundary, or when the three detectors disagree sharply with each other.
4. **Adjust the governance sliders** — change the medium/high risk thresholds per use case live and watch the same detection scores produce different tiered actions.
5. **Override a decision** — every row in the audit trail can be overridden with a reason, simulating the human-in-the-loop feedback signal a production system would use to recalibrate.
6. **Watch the feedback loop close itself** — override two similar flagged decisions the same way (e.g. "Dismiss flag" twice on the same use case), then look at panel 2: the relevant threshold slider moves on its own, and panel 7 logs why. This turns "overrides feed the improvement loop" from a caption into an actual mechanism.
7. **Set downstream automated actions** — the slider under the response box simulates an agentic pipeline (e.g. this response triggers an email send or a record update). With actions > 0, the decision card shows a compounded exposure estimate, making multi-turn/agentic risk visible rather than treating every response as independent.
8. **Switch jurisdiction** — toggling US / EU changes how hard PII hits are weighted (GDPR context is stricter), standing in for a jurisdiction-aware policy layer.

Why this design

The brief's real-world complexities directly shaped the architecture:

Complexity from the brief	How the prototype addresses it
Different use cases have different risk/latency budgets	Each use case has its own detection weights and thresholds (<code>USE_CASES</code> config) — a customer chatbot weighs Responsibility highest, a regulated decision-support tool weighs Performance highest with stricter thresholds
Bias / hallucination / privacy risks overlap	The Responsibility and Performance scores are computed independently but combined into one weighted overall score, rather than forcing a single category label
No reliable real-time ground truth	The grounding checker degrades gracefully: with a source doc it scores lexical overlap; without one, it falls back to a claim-assertiveness heuristic (flags absolute/unhedged language) and says so explicitly in its output
Communicating uncertainty, not just a score	Every check reports a confidence percentage alongside its risk tier — separate from the risk level itself. Confidence drops when there's no source to verify against, when the score sits right on a decision-tier boundary, or when the three detectors disagree sharply
Over-flagging vs under-flagging tradeoff	The "Alert-Fatigue Tradeoff Simulator" sweeps the flagging threshold across the batch results and plots false positives against false negatives, making the tuning tradeoff visible and explicit rather than solving it away
Enterprises consume models via API, not owning them	All checks work purely on the <i>input/output text</i> , never on model internals — this is designed to sit outside the model as middleware
Regulatory / policy differences by geography and use case	The threshold sliders are a stand-in for a configurable policy layer — same detectors, different governance per deployment. A jurisdiction toggle (US / EU) additionally re-weights PII detections stricter under a GDPR context
Feedback loops	The audit trail's override mechanism logs every human correction. This is no longer just a caption: repeated "Dismiss flag" overrides at a use case + tier automatically raise that threshold, and repeated "Escalate further" overrides lower it — visible live in the governance sliders and logged in the new "Feedback Loop & Recalibration" panel
False positive / negative rate reporting to a skeptical stakeholder	The same override log is reused to compute a live "estimated false-positive rate of flagged responses" stat (dismissed overrides ÷ total flagged), alongside the batch-level alert-fatigue tradeoff chart
Multi-turn / agentic use cases compound risk across turns	A "downstream automated actions" control lets a response be marked as feeding N further automated steps; the decision card then shows a compounded exposure estimate reflecting that one bad output can propagate

Complexity from the brief

How the prototype addresses it

Bias / hallucination / privacy risks overlap in practice

When a response trips both the Performance and Responsibility detectors past a threshold, the UI now surfaces an explicit "overlapping risk" note rather than silently picking one label, making the scoring philosophy (joint weighting, not single-category classification) visible in the demo itself

Architecture

flowchart LR
A[AI Model Response] --> B{Parallel Fast Classifiers}
B --> C[Performance Check
grounding overlap /
assertiveness heuristic]
B --> D[Cost Check
token volume vs
use-case baseline]
B --> E[Responsibility Check
PII regex +
policy phrase match]
C --> F[Weighted Risk Score]
D --> F
E --> F
F --> G{Tiered Decision
per use-case policy}
G -->|Low| H[Log silently, allow]
G -->|Medium| I[Auto-redact / flag for review]
G -->|High| J[Block, escalate to human]
G -->|Borderline| K[Heavier AI-as-judge
verifier pass]
I --> L[Audit Trail]
J --> L
H --> L
L --> M[Human Override]
M --> N[Feedback Loop →
recalibrate thresholds]

In production, "Parallel Fast Classifiers" would be actual lightweight models/embeddings running asynchronously (adding milliseconds, not seconds), with the heavier verifier LLM-as-judge pass reserved only for scores that land in the borderline band — exactly as scoped in the concept deck. The prototype simulates this same shape deterministically so the mechanism is inspectable and reproducible without needing live model calls or API keys.

What's simulated vs. what would change in production

Being transparent about scope, since this is a proof-of-concept on illustrative data, not a production system:

- **Grounding check:** prototype uses lexical word-overlap; production would use embedding similarity + retrieval verification against a real knowledge base, plus an actual LLM-as-judge pass for borderline cases.
- **PII / policy detection:** prototype uses regex + a small keyword list; production would use dedicated PII/entity-detection models and a maintained, regularly-updated policy ruleset per jurisdiction.
- **Cost anomaly detection:** prototype compares against a fixed baseline; production would track a rolling statistical baseline per use case and flag genuine anomalies (z-score / EWMA), including retries and rework, not just single-response length.
- **Scale:** prototype demonstrates the mechanism on 12 representative interactions; the reference parameters assume tens of thousands of weekly interactions, which the architecture is designed to support since each check is a lightweight, parallelizable, stateless function.

Assumptions

- Enterprise runs 3 concurrent AI use cases (customer-facing, internal, regulated decision-support), each with distinct latency/risk tolerance, per the reference parameters in the brief.

- Foundation models are consumed via API, so the checker never inspects model weights/internals — only input/output text (input/output-layer middleware).
- "Truly risky" labels used in the batch demo's tradeoff chart are our own annotations for demonstration purposes, standing in for the labeled incident data a real deployment would accumulate over time.

Repository structure

```
/index.html    — the working prototype (single file, no build step)
/README.md     — this file
```

Team / Submission

Team Muffin — Accenture Innovation Challenge 2026 — Round 2 Prototype, Problem Track 1: ControlPlane.ai

